EPIDEMIC CONTROL

Using Particle Swarm Optimization

Optimizing city lockdown policies to minimize infections while protecting the economy

THE PROBLEM

SCENARIO

- A COVID-like outbreak is spreading across a city with 8 distinct zones
- Each zone has different population, risk level, and economic importance
- The government must decide: how strictly to restrict each zone?

Too strict → economy collapses

Too lenient → hospitals overwhelmed

WHY PSO?

✗ **Brute Force**

Infinite combinations — impossible

✗ **Genetic Algorithm**

Designed for discrete problems

✓ **PSO**

Built for continuous real-valued search spaces

The optimal policy is not obvious — PSO tries to finds it.

CITY ZONES

Problem Setup

Zone	Population	Risk	Econ Val	Notes
City Center	10,000	1.5	1.5	*High density, offices, transport*
North District	8,000	1.0	1.0	*Average residential area*
East District	12,000	1.1	0.9	*Large suburb, moderate risk*
South District	6,000	0.9	0.8	*Lower density, quieter area*
West District	9,000	1.0	1.1	*Mixed commercial/residential*
University Area	15,000	1.7	1.2	*High mixing, student dorms*
Industrial Zone	7,000	0.8	2.0	*Factories — very expensive!*
Suburb North	5,000	0.6	0.5	*Low density, low risk*

Risk: High Medium Low

STEP 1 — Representation

Each particle = one complete city-wide restriction policy

72%	45%	61%	38%	50%	80%	25%	42%
City Center	North Dist.	East Dist.	South Dist.	West Dist.	Univ. Area	Indust. Zone	Suburb North

velocity (starts at 0.0 — particles are stationary before first move):

```
[ 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0 ]
```

STEP 2 — Fitness Function

```
fitness = ( 0.5 × norm_infected + 0.5 × norm_cost )
```

① Effective R

eff_R = R0 × risk × (1 − restriction)
R0=2.5 is real COVID-19 value.
risk adjusts per zone density.
(1−restriction) = fraction of contacts remaining.

② Attack Rate

attack_rate = 1 − (1/eff_R) if eff_R > 1
Fraction of population eventually infected.
From real epidemiology (herd immunity formula).
If eff_R ≤ 1 → disease dying out → rate = 0.

③ Residual Spread

infected = pop × (attack_rate + 0.02)
0.02 = 2% always infected regardless.
Essential workers, household transmission.
Mirrors real COVID — zero infections impossible.

④ Economic Cost

cost = restriction × pop × 100 × econ_val
100 = Money lost per person per 100% restriction.
econ_val = 2 for Industrial Zone → 2× more expensive.

PSO looks for the minimum fitness → fewer infections + less damage = better policy

STEP 3 — Velocity & Position Update

```
v_new = w·v_old + c1·r1·(pbest − pos) + c2·r2·(gbest − pos)
```

→ Inertia

```
w × v_old    (w = 0.5)
```

- Keeps particle moving in the same direction. Resistance to movement.
- Higher inertia (0.8–1) → more exploration
- Lower inertia (0.1–0.4) → more exploitation
- Ours (0.5) balances both

↖ Cognitive

```
c1 × r1 × (pbest − pos)    (c1 = 1.5)
```

- Pulls particle back toward its own personal best. Promotes individual exploration.
- Cognitive constant between 0 and 2
- Higher c1 = more individual exploration

★ Social

```
c2 × r2 × (gbest − pos)    (c2 = 1.5)
```

- Pulls particle toward the swarm's global best.
- Social acceleration is determined by using a constant and scaling it with a random number.
- R2 encourages diversity in favoring the social factor.

```
position_new = position_old + velocity_new    then clip to [0.0, 1.0]
```

STEP 4 — pbest, gbest & Convergence

pbest — Personal Best

Each particle remembers the best position it has ever personally visited.

If new fitness < pbest fitness → update pbest

gbest — Global Best

The single best position found by ANY particle in the entire swarm ever.

If any particle improves → gbest updates → ALL particles immediately pull toward it.
This is the "swarm intelligence" — collective knowledge sharing.

Convergence — When Does PSO Stop?

```
if particle.fit  <  gbest_fit + 1e-7:    meaningful improvement → reset stall counter  (+ 1e-7 to ignore small fractional changes)
```

```
if stall_count  >=  40:    no improvement for 40 consecutive iterations  →  STOP
```

Iteration 0 | fitness: 0.384923 ← *starts bad (particles near extremes)*

Iteration 20 | fitness: 0.331232 ← *improving, swarm converging*

RESULTS — Optimal Policy Found

💡 **Key Insight: University Area (risk 1.5, average loss) → highest restriction. Industrial Zone (risk 0.8, very expensive) → no restriction.**

ZONE	ORIGINAL POP	RESTR.	INFECTED	ECON COST
City Center	10,000	73.3%	200	1100001
North District	8,000	60.0%	160	480000
East District	12,000	63.6%	240	687274
South District	6,000	55.6%	120	266667
West District	9,000	60.0%	180	594000
University Area	15,000	76.5%	300	1376471
Industrial Zone	7,000	0.0%	3640	0
Suburb North	5,000	33.3%	100	83334
TOTAL	72,000	-	4,940	4,587,745

PSO SUMMARY

Epidemic Control — Key Takeaways

01 **Representation:** Each particle = one 8-zone restriction policy (real values 0–1)

02 **Fitness:** Minimize infections + cost using eff_R, attack rate, residual spread

03 **Velocity:** w=0.5, c1=1.5, c2=1.5

04 **pbest & gbest:** Personal memory + swarm knowledge guide all 50 particles every iteration

05 **Convergence:** Stops after 40 iterations of no meaningful improvement (stall limit)

PSO efficiently finds the optimal policy without checking all combinations — this is its core advantage over brute force search